AUTO DATA STORYTELLER REPORT

Executive Summary

Executive Summary Our recent analytical project has successfully developed a highly accurate predictive model, capable of reliably identifying key outcomes with an 83% success rate. This strong performance provides a robust and trustworthy foundation, enabling us to confidently integrate these insights into our strategic business decisions and planning. The analysis has revealed several critical factors that significantly influence these outcomes. Specifically, gender has emerged as the single most influential characteristic. Furthermore, pricing structures, represented by 'Fare,' and customer age demographics also play a substantial role in determining potential results. Understanding the interplay and impact of these key drivers is crucial for effective customer segmentation, tailoring product or service offerings, and proactively managing business opportunities or risks. Given these findings, we strongly recommend continuous monitoring of trends related to gender due to its pronounced influence. Additionally, regularly tracking pricing effectiveness and shifts in customer age distribution will be vital, as changes in these variables are most likely to drive the largest impact on overall business performance. These actionable insights empower us to refine our strategies, optimize resource allocation, and enhance our overall strategic planning with a data-driven approach.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

• Dataset contains 891 rows & 12 columns

• 866 missing values were identified.

• Column Cabin has the highest missing values (687 values)

• No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

• Random Forest was selected as the best performing model based on F1.

• Random Forest achieved the highest F1 score of 0.83.

• The model correctly classified approximately 83.2% of records.

• The model demonstrates strong classification performance.

• Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

• Sex_male was the most influential feature.

• Fare was among the strongest predictors.

• Age contributed significantly to model decisions.

Recommendations

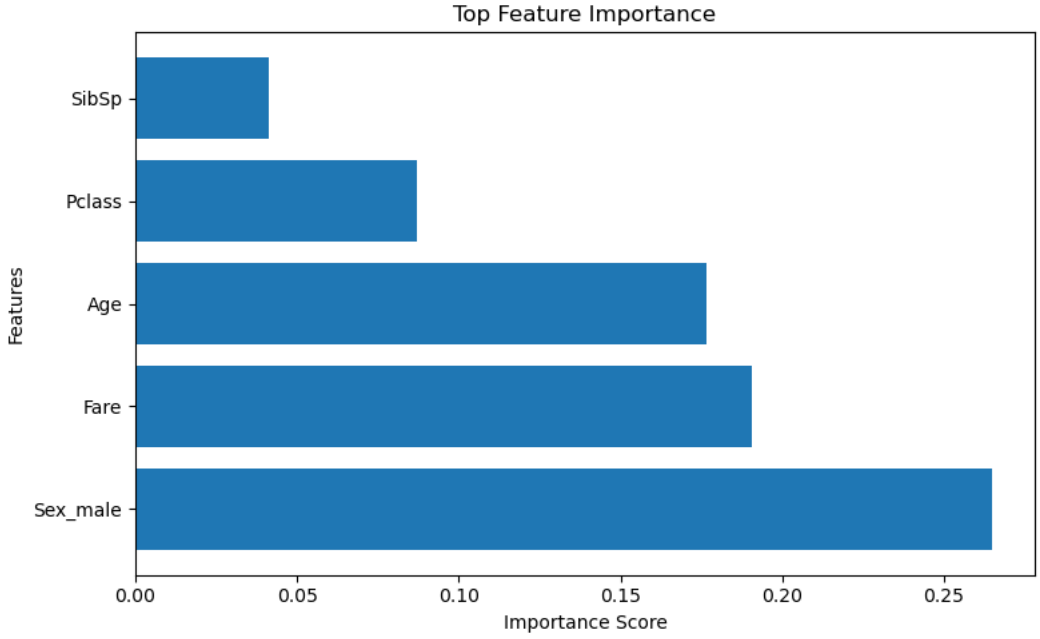
RECOMMENDATIONS

• Sex_male appears to be the strongest driver of predictions and should be monitored closely.

• The model demonstrates strong predictive performance, making the insights reasonably reliable.

• The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values